Detailed Solutions to Practice Problems on LAD Regression

Advanced Statistical Methods II

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Abstract

This document gives self-contained, step-by-step solutions to all fifteen problems in the attached Practice Problems on LAD Regression sheet. Each original question is reproduced in full before the solution. The theory required for that particular problem is then stated explicitly, and every important algebraic, optimization, and conceptual transition is explained. The notation and the KKT, uniqueness, conditional-median, and weighted-median tools follow the supplied problem sheet. The discussion of why absolute loss is useful and how it can be optimized by majorization follows the attached ASM II lecture notes.

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Preliminary theory: derivation of the four useful LAD results

Before solving the fifteen questions, we derive the four tools stated on the practice sheet. They are used repeatedly below, so the goal is to understand not only their formulas but also why they are true.

Tool A: KKT (subgradient) conditions for LAD

For a line $a + bx$, write

$$r_i(a,b) = y_i - a - bx_i, \qquad Q(a,b) = \sum_{i=1}^{n} |r_i(a,b)|.$$

The absolute-value function is convex. Its subgradient at u is

$$\partial|u| = \begin{cases} \{1\}, & u > 0, \\ [-1,1], & u = 0, \\ \{-1\}, & u < 0. \end{cases}$$

Choose $s_i \in \partial|r_i|$. Because the derivative of r_i with respect to (a,b) is $(-1,-x_i)$, the subgradient contribution of observation i to Q is

$$-s_i \begin{pmatrix} 1 \\ x_i \end{pmatrix}.$$

For a convex function, a point is a global minimizer if and only if zero belongs to its subgradient. Hence

$$\sum_{i=1}^{n} s_i = 0, \qquad \sum_{i=1}^{n} x_i s_i = 0,$$

where $s_i = \operatorname{sign}(r_i)$ for nonzero residuals and $s_i \in [-1,1]$ for zero residuals. This is exactly the certificate condition in the problem sheet. If the intercept is omitted, only the slope equation remains:

$$\sum_{i=1}^{n} x_i s_i = 0.$$

Tool B: a sufficient condition for uniqueness

Suppose (a,b) is a minimizer with certificate (s_i). For any competing line (a',b'), let $r_i' = y_i - a' - b'x_i$. The subgradient inequality for absolute value gives

$$|r_i'| \geq |r_i| + s_i(r_i' - r_i).$$

Summing over observations,

$$\begin{aligned} Q(a',b') &\geq Q(a,b) + \sum_i s_i\{-(a'-a) - (b'-b)x_i\} \\ &= Q(a,b) - (a'-a)\sum_i s_i - (b'-b)\sum_i x_i s_i \\ &= Q(a,b), \end{aligned}$$

where the final equality uses the two KKT equations. Now suppose $r_j = r_k = 0$, $x_j \neq x_k$, and $|s_j| < 1$, $|s_k| < 1$. At either such zero residual, the inequality

$$|u| \geq s_i u$$

is strict for every $u \neq 0$ when $|s_i| < 1$. If (a', b') is also a minimizer, equality must hold in the summed inequality, so it must have $r'_j = r'_k = 0$. Thus

$$a' + b'x_j = a + bx_j, \qquad a' + b'x_k = a + bx_k.$$

Subtracting these equations and using $x_j \neq x_k$ gives $b' = b$, and then $a' = a$. Therefore the minimizer is unique.

Tool C: a linear conditional median minimizes population LAD

Assume $\mathbb{E}|Y| < \infty$ and that $m(X) = a_0 + b_0X$ is a conditional median of Y given X almost surely. For any fixed value x, a median minimizes expected absolute deviation, so for every real number c,

$$\mathbb{E}(|Y - c| \mid X = x) \geq \mathbb{E}(|Y - m(x)| \mid X = x).$$

Set $c = a + bx$ and take expectations over X. The result is

$$\mathbb{E}|Y - a - bX| \geq \mathbb{E}|Y - a_0 - b_0X|.$$

Hence (a_0, b_0) minimizes the population LAD objective.

The analogous OLS statement follows from the conditional squared-error decomposition. If $\mathbb{E}(Y \mid X) = a_0 + b_0X$ and $\mathbb{E}(Y^2) < \infty$, then

$$\mathbb{E}\{(Y - c)^2 \mid X\} = \mathrm{Var}(Y \mid X) + \{\mathbb{E}(Y \mid X) - c\}^2.$$

The first term does not depend on c, and the second is minimized at $c = a_0 + b_0X$. Taking $c = a + bX$ and averaging proves that (a_0, b_0) is the population OLS target.

Tool D: weighted medians

Let

$$g(b) = \sum_i w_i|z_i - b|, \qquad w_i > 0, \qquad W = \sum_i w_i.$$

For $z_i < b$, the derivative contribution with respect to b is $+w_i$; for $z_i > b$, it is $-w_i$; and for $z_i = b$, it is any number in $[-w_i, w_i]$. Let

$$L = W(z < b), \quad E = W(z = b), \quad R = W(z > b), \quad L + E + R = W.$$

A zero subgradient can be selected exactly when

$$|L - R| \leq E.$$

This is equivalent to $L \leq W/2$ and $R \leq W/2$. Since $W(z \leq b) = L + E = W - R$, the same condition can be written

$$\boxed{W(z < b) \leq \frac{W}{2} \leq W(z \leq b).}$$

This is the weighted-median condition stated on the sheet. Taking all weights equal gives the usual sample-median characterization.

Connection to Lecture VII: MM computation of LAD

The supplied ASM II notes explain why absolute loss is useful: it is less sensitive to large residuals than squared loss, but is not differentiable at zero residuals (Lecture VII, pages 274–277). They construct a smooth majorizer from

$$0 \leq (|u| - |u^{(t)}|)^2,$$

which rearranges, for $u^{(t)} \neq 0$, to

$$|u| \leq \frac{u^2}{2|u^{(t)}|} + \frac{|u^{(t)}|}{2}.$$

For LAD residuals $r_i(\beta) = y_i - x_i^T\beta$, this gives a majorizer whose parameter-dependent part is

$$\sum_i \frac{\{y_i - x_i^T\beta\}^2}{2|r_i(\beta^{(t)})|}.$$

Consequently, minimizing the majorizer is a weighted least-squares step with

$$w_i^{(t)} = \frac{1}{2|r_i(\beta^{(t)})|}, \qquad \beta^{(t+1)} = (X^T W^{(t)} X)^{-1} X^T W^{(t)} y,$$

when the inverse exists. The lecture notes recommend replacing the denominator near zero by a small positive safeguard such as $|r_i^{(t)}| + \varepsilon$. The KKT certificate proves exact optimality; the MM construction is an iterative computational route toward an LAD solution.

1 Problem 1: Existence and geometry of LAD minimizers

Problem Statement

Suppose not all x_i's are equal. Consider LAD regression of y on x with intercept.

(a) Show that a minimiser of $Q(a, b)$ exists.

(b) Show that there is some LAD line which passes through at least two data points (x_j, y_j) and (x_k, y_k) with $x_j \neq x_k$.

(c) Must every LAD line pass through two data points?

Required Theory

For observations (x_i, y_i), the LAD objective is

$$Q(a, b) = \sum_{i=1}^{n} |y_i - a - bx_i| = \sum_{i=1}^{n} |r_i(a, b)|.$$

It is continuous and convex because absolute value is continuous and convex and each residual is affine in (a, b). The design matrix with rows $(1, x_i)$ has rank two exactly when the x_i are not all equal.

The KKT certificate supplied in the problem sheet says that (a, b) is an LAD minimizer if and only if there are s_i such that

$$s_i = \operatorname{sign}(r_i) \quad (r_i \neq 0), \qquad s_i \in [-1, 1] \quad (r_i = 0),$$

with

$$\sum_i s_i = 0, \qquad \sum_i x_i s_i = 0.$$

A zero residual means that the fitted line passes through the corresponding observation. A useful uniqueness condition from the sheet is: if two zero residuals occur at distinct predictor values and their certificate values satisfy $|s_j| < 1$ and $|s_k| < 1$, the LAD solution is unique.

Solution

(a) Existence of a minimizer.

Step 1: Write the objective in matrix form. Let

$$A = \begin{pmatrix} 1 & x_1 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix}, \qquad \gamma = \begin{pmatrix} a \\ b \end{pmatrix}, \qquad y = (y_1, \ldots, y_n)^T.$$

Then

$$Q(a, b) = \|y - A\gamma\|_1.$$

Step 2: Use the assumption on the predictors. Since the x_i are not all equal, the two columns of A are linearly independent. Thus $A\gamma = 0$ implies $\gamma = 0$.

Step 3: Show that the objective becomes large when the coefficients become large. On the unit circle $\{u \in \mathbb{R}^2 : \|u\|_2 = 1\}$, the continuous function $u \mapsto \|Au\|_1$ is strictly positive. The unit circle is compact, so it has a positive minimum; call it $c > 0$. By homogeneity,

$$\|A\gamma\|_1 \geq c\|\gamma\|_2 \quad \text{for every } \gamma \in \mathbb{R}^2.$$

The triangle inequality gives

$$Q(a,b) = \|y - A\gamma\|_1 \geq \|A\gamma\|_1 - \|y\|_1 \geq c\|\gamma\|_2 - \|y\|_1.$$

The right-hand side tends to infinity as $\|(a,b)\|_2 \to \infty$. Hence Q is coercive.

Step 4: Apply the minimum principle. Choose a sufficiently large closed ball outside which $Q(a,b) > Q(0,0)$. A global minimizer must lie inside that ball. The ball is compact and Q is continuous, so Q attains its minimum there. Therefore a global LAD minimizer exists.

(b) At least one minimizer passes through two observations at distinct x-values.

The objective is piecewise affine in (a,b). The lines

$$y_i - a - bx_i = 0, \qquad i = 1, \ldots, n,$$

partition the coefficient plane into regions. Inside any one open region the signs of all residuals are fixed, so Q is an affine function of (a,b).

Step 1: Move a minimizer to the boundary of one such region. If a minimizer has no zero residual, it lies in an open region on which Q is affine. An affine function that has a local minimum at an interior point must be constant in a neighbourhood of that point. We may therefore move along a line segment, without changing the objective, until at least one residual becomes zero. The endpoint remains a minimizer.

Step 2: If all zero-residual observations have the same x-value, move while preserving them. Suppose the current minimizer has zero residuals only at predictor value x_0. The coefficient direction

$$(a,b) \longmapsto (a - tx_0, b + t)$$

leaves $a + bx_0$ unchanged. Hence it preserves every zero residual at x_0. Along a sufficiently short segment in this direction, the signs of all nonzero residuals remain unchanged. As before, the objective is affine along this segment and has a minimum at its interior point; it is therefore constant along the segment.

Step 3: Continue until a second, distinct predictor value is hit. Since the x_i are not all equal, there is an observation with $x_i \neq x_0$. Along the direction above, its fitted value changes at rate $x_i - x_0 \neq 0$. Moving in the direction in which that residual approaches zero, and stopping at the first residual-zero point, preserves the objective and creates a zero residual at an x-value different from x_0. The resulting line is still an LAD line and passes through two observations with distinct predictor values.

This proves the existence of at least one such LAD line. Notice that the argument selects a particular minimizer; it does not claim that every minimizer has this property.

(c) Not every LAD line has to pass through two data points.

Consider the four observations

$$(0,0), \quad (0,1), \quad (1,0), \quad (1,1).$$

For a candidate line $y = a + bx$, its objective is

$$Q(a,b) = |a| + |1 - a| + |a + b| + |1 - a - b|.$$

For every real number z, $|z| + |1 - z| \geq 1$, with equality exactly when $z \in [0,1]$. Applying this fact to $z = a$ and $z = a + b$ gives

$$Q(a,b) \geq 2,$$

with equality precisely when

$$0 \leq a \leq 1, \qquad 0 \leq a + b \leq 1.$$

In particular, $(a,b) = (1/2, 0)$ is an LAD solution. Its fitted value is $1/2$ at both $x = 0$ and $x = 1$, so it passes through none of the four observations. Thus the answer is no.

Key Takeaway

There is always at least one LAD minimizer supported by two observations at distinct predictor values, but non-uniqueness can produce other optimal lines that do not pass through two data points.

2 Problem 2: Pooling two subpopulations with identical LAD fits

Problem Statement

Two disjoint subpopulations S_1 and S_2 are given. We fit LAD with intercept of y on x separately in each. Suppose both fits give the same optimal coefficients (α^*, β^*). Now regress y on x by LAD on $S_1 \cup S_2$.

(a) Is (α^*, β^*) an LAD solution for the union?

(b) Must the LAD estimates for the union be (α^*, β^*)?

(c) What if we consider the same question for OLS?

Required Theory

Write the two objective functions as $Q_1(\alpha, \beta)$ and $Q_2(\alpha, \beta)$. Because the samples are disjoint, the pooled LAD objective is their sum:

$$Q_{\text{pool}}(\alpha, \beta) = Q_1(\alpha, \beta) + Q_2(\alpha, \beta).$$

If (α^*, β^*) minimizes each term separately, then for every (a, b),

$$Q_1(a, b) \geq Q_1(\alpha^*, \beta^*), \qquad Q_2(a, b) \geq Q_2(\alpha^*, \beta^*).$$

Adding these inequalities proves pooled optimality. Equality need not identify a unique point if either objective has a flat set of minimizers.

For OLS, a fitted coefficient vector with intercept satisfies the two normal equations

$$\sum_i e_i = 0, \qquad \sum_i x_i e_i = 0.$$

The pooled sums are sums of the corresponding group sums. Therefore coefficients satisfying the normal equations in both groups satisfy them in the union.

Solution

(a) Yes.

Let $m_g = \min_{(a,b)} Q_g(a, b)$ for $g = 1, 2$. The assumption says

$$Q_1(\alpha^*, \beta^*) = m_1, \qquad Q_2(\alpha^*, \beta^*) = m_2.$$

For any candidate (a, b),

$$Q_{\text{pool}}(a, b) = Q_1(a, b) + Q_2(a, b) \geq m_1 + m_2.$$

At (α^*, β^*) the pooled objective equals $m_1 + m_2$. Hence it is a pooled LAD minimizer.

(b) It need not be the only pooled estimate.

Use the unit-square data from Problem 1 for each subpopulation:

$$S_1 = S_2 = \{(0,0),(0,1),(1,0),(1,1)\}.$$

For either group, the complete set of LAD solutions is

$$0 \le a \le 1, \qquad 0 \le a + b \le 1.$$

Thus $(\alpha^*, \beta^*) = (1/2, 0)$ is an optimal fit in both groups. On pooling the groups, the objective is simply twice the original objective. Its minimizer set is unchanged, so the pooled fit is not unique. A computational routine may return a different point in that same minimizer set. The correct statement is that (α^*, β^*) remains an LAD solution, not that it must be the uniquely selected coefficient pair.

(c) OLS.

The same aggregation argument applies to the OLS normal equations. If (α^*, β^*) is the OLS coefficient vector in each group, then in the union

$$\sum_{i \in S_1 \cup S_2} e_i = \sum_{i \in S_1} e_i + \sum_{i \in S_2} e_i = 0,$$

and similarly $\sum_{S_1 \cup S_2} x_i e_i = 0$. Thus it is a pooled OLS solution. If the pooled predictor values are not all equal, the pooled design matrix has full column rank and the OLS coefficients are unique; in that usual case the pooled OLS estimate must be (α^*, β^*).

Key Takeaway

For both losses, common groupwise optima remain pooled optima. The important difference is that LAD can have a flat set of minimizers, whereas OLS with a full-rank intercept design has a unique coefficient vector.

3 Problem 3: Pooling groups with a common intercept and different positive slopes

Problem Statement

> Consider the same setup as in Problem 2. Now suppose that in the two subpopulations the LAD fits are
>
> $$(\alpha, \beta_1) \quad \text{and} \quad (\alpha, \beta_2), \qquad \beta_1 \neq \beta_2, \quad \beta_1, \beta_2 > 0,$$
>
> i.e. the same intercept but different slopes. We fit LAD on the union.
>
> (a) Does the intercept remain equal to α?
>
> (b) Does the pooled slope lie between β_1 and β_2?

Required Theory

The pooled objective is still $Q_1(a,b) + Q_2(a,b)$. Convexity of each group objective does not by itself imply that the minimizer of the sum is obtained by averaging the groupwise minimizers, nor that either coordinate of the minimizer remains in a coordinatewise interval. The KKT

certificate is the decisive check: a proposed pooled line is optimal if one can choose subgradient values at its zero residuals so that both the intercept and slope equations hold.

For a three-point group with two zero residuals at $x_j \neq x_k$ and one positive residual, choosing the nonzero residual's certificate to be $+1$ and solving the two KKT equations for the zero residuals gives a certificate. If both zero-residual certificate values have absolute value strictly less than one, the supplied uniqueness tool applies.

Solution

Both answers are no. We give explicit data and verify all claimed fits rather than relying on a picture.

(a) The pooled intercept need not remain α.

Take $\alpha = 0$, $\beta_1 = 1$, and $\beta_2 = 2$. Let

	x	y	groupwise fitted value
S_1	-1	-1	-1
S_1	0	1	0
S_1	1	1	1
S_2	-1	-2	-2
S_2	0	1	0
S_2	1	2	2

For S_1, the line $y = x$ has residuals $(0, 1, 0)$. A certificate is

$$(s_{-1}, s_0, s_1) = (-1/2, 1, -1/2),$$

which satisfies $\sum s_i = 0$ and $\sum x_i s_i = 0$. The two zero residuals have strict certificate values, so the fit is unique. Exactly the same certificate proves that $y = 2x$ is the unique LAD fit in S_2.

For the pooled sample, consider

$$\widehat{y} = \frac{1}{2} + \frac{3}{2}x.$$

The residuals, in the order in which the observations are listed, are

$$r_{S_1} = (0, 1/2, -1), \qquad r_{S_2} = (-1, 1/2, 0).$$

Thus the nonzero signs sum to zero and their x-weighted sum is also zero. Set the two zero-residual certificate values equal to zero. The KKT equations hold, so the pooled line is an LAD minimizer. It has intercept $1/2$, not 0. The two strict zero-residual certificate values are both zero and occur at distinct x-values, so the pooled minimizer is unique.

(b) The pooled slope need not lie between the two group slopes.

Again take the groupwise lines to be $y = x$ and $y = 2x$, with the same intercept zero. Use the following samples:

group	x	y
S_1	-3	-3
S_1	-2	0
S_1	3	3
S_2	-2	-4
S_2	-1	2
S_2	1	2

For S_1, the line $y = x$ has residuals $(0, 2, 0)$. A certificate is

$$(s_{-3}, s_{-2}, s_3) = (-5/6, 1, -1/6),$$

which satisfies both KKT equations and is strict at the two zero residuals. For S_2, the line $y = 2x$ has residuals $(0, 4, 0)$ and certificate

$$(s_{-2}, s_{-1}, s_1) = (-2/3, 1, -1/3).$$

Consequently, the individual fits are unique and have slopes 1 and 2.

Now consider the pooled line

$$\widehat{y} = \frac{4}{3} + \frac{2}{3}x.$$

Its residuals in the displayed order are

$$\left(-\frac{7}{3},\ 0,\ -\frac{1}{3},\ -4,\ \frac{4}{3},\ 0\right).$$

Choose certificate values equal to the residual signs at nonzero residuals and set the two zero-residual certificate values to 1 each. Then

$$\sum_i s_i = (-1) + 1 + (-1) + (-1) + 1 + 1 = 0,$$

and

$$\sum_i x_i s_i = 3 - 2 - 3 + 2 - 1 + 1 = 0.$$

Thus the line satisfies the KKT conditions and is a pooled LAD fit. Its slope is $2/3$, which is smaller than both 1 and 2. Hence the pooled slope need not lie between the group-specific slopes.

Key Takeaway

Even when the groupwise intercepts agree and both group slopes are positive, pooling can change the intercept and can produce a slope outside the interval between the two group slopes. The pooled objective must be solved as its own optimization problem.

4 Problem 4: Pooling groups with ordered intercepts and slopes

Problem Statement

> Consider the same setup as in Problem 2, but now the subpopulation LAD fits are (α_1, β_1) and (α_2, β_2) with
>
> $$0 < \alpha_1 < \alpha_2, \qquad 0 < \beta_1 < \beta_2.$$
>
> (a) Must the pooled LAD estimates satisfy $\widehat{\alpha} \in [\alpha_1, \alpha_2]$ and $\widehat{\beta} \in [\beta_1, \beta_2]$?
>
> (b) Must $\widehat{\beta}$ at least be positive?
>
> (c) Is there a natural condition under which the pooled estimates are in the intervals?

Required Theory

The separate LAD minimizers constrain each group's own objective, but do not impose coordinatewise bounds on the minimizer of their sum. The KKT conditions can be checked by constructing subgradient certificates. A particularly useful fact from the problem sheet is that

a certificate with two zero residuals at distinct x-values and strict subgradient values at those residuals proves uniqueness.

At the population level, the supplied conditional-median result says that if the pooled conditional median is an affine function $m(x) = a + bx$ and $\mathbb{E}|Y| < \infty$, then (a, b) minimizes the pooled expected absolute loss. Thus a sufficient condition for interval conclusions is that the pooled conditional median itself is affine and has coefficients in the stated intervals. Groupwise affine medians alone do not guarantee this property.

Solution

(a) No. We construct a pooled fit with both coordinates outside the intervals.

Let the groupwise LAD lines be

$$(\alpha_1, \beta_1) = (0, 1), \qquad (\alpha_2, \beta_2) = (1, 2).$$

Take the following data:

group	x	y
S_1	-5	-5
S_1	-4	-3
S_1	-3	-3
S_2	-5	-9
S_2	-4	-6
S_2	-2	-3

The first group has line $y = x$, with residuals $(0, 1, 0)$ and certificate

$$(-1/2, 1, -1/2).$$

The second group has line $y = 1 + 2x$, also with residuals $(0, 1, 0)$ in its own row order, and certificate

$$(-2/3, 1, -1/3).$$

Both groupwise fits are unique by the supplied uniqueness criterion.

For the union, consider

$$\widehat{y} = -\frac{5}{3} + \frac{2}{3}x.$$

The six residuals are

$$\left(0, \frac{4}{3}, \frac{2}{3}, \ -4, -\frac{5}{3}, 0\right).$$

At the nonzero residuals the certificate values are $(+1, +1, -1, -1)$. At the zero residuals (which occur at $x = -5$ in S_1 and $x = -2$ in S_2), choose respectively $2/3$ and $-2/3$. Then

$$\sum_i s_i = 1 + 1 - 1 - 1 + \frac{2}{3} - \frac{2}{3} = 0,$$

and

$$\sum_i x_i s_i = (-4) + (-3) + (5) + (4) - \frac{10}{3} + \frac{4}{3} = 0.$$

Therefore this is a pooled LAD fit. It has

$$\widehat{\alpha} = -\frac{5}{3} < 0 = \alpha_1, \qquad \widehat{\beta} = \frac{2}{3} < 1 = \beta_1,$$

so both interval assertions fail. The two zero-residual certificate values have absolute value less than one, and the corresponding x-values differ; hence the pooled fit is unique.

(b) No. A pooled slope can be negative even when both groupwise slopes are positive.

Take

$$(\alpha_1, \beta_1) = (0, 1), \qquad (\alpha_2, \beta_2) = (20, 2),$$

with exact two-point samples

$$S_1 : \ (10, 10), (11, 11), \qquad S_2 : \ (0, 20), (1, 22).$$

Each group's two points determine its stated line exactly, so each groupwise LAD fit is unique. For the pooled sample, consider

$$\widehat{y} = 20 - \frac{9}{11}x.$$

The residuals at $(0, 20)$ and $(11, 11)$ are zero. The residual at $(1, 22)$ is $31/11 > 0$, and the residual at $(10, 10)$ is $-20/11 < 0$. Assign certificate values $s_1 = -9/11$ at $x = 0$ and $s_4 = 9/11$ at $x = 11$; the nonzero residuals have certificate values $+1$ at $x = 1$ and -1 at $x = 10$. Then

$$\sum_i s_i = -\frac{9}{11} + 1 - 1 + \frac{9}{11} = 0,$$

while

$$\sum_i x_i s_i = 0 + 1 - 10 + 11\frac{9}{11} = 0.$$

The KKT certificate is valid and is strict at the two zero residuals, so the line is the unique pooled LAD fit. Its slope is $-9/11 < 0$. Thus positivity is not preserved by pooling.

(c) A natural sufficient condition.

A clean population-level sufficient condition is the following: the group membership is independent of X, and the pooled conditional median is itself affine,

$$\operatorname{med}(Y \mid X = x) = a_0 + b_0 x, \qquad a_0 \in [\alpha_1, \alpha_2], \quad b_0 \in [\beta_1, \beta_2].$$

Then the conditional-median result from the problem sheet immediately implies that (a_0, b_0) is a pooled LAD target, and it lies in the requested intervals. This condition is stronger than merely assuming that each group's conditional median is linear. It explicitly requires the mixture of the two conditional distributions to have a linear median with interval-valued coefficients.

One transparent special case is a common slope, $\beta_1 = \beta_2 = \beta$, with group-independent X and group-specific location shifts plus a common error distribution whose mixture has a fixed median shift $c \in [\alpha_1, \alpha_2]$ after accounting for the common slope. The pooled conditional median is then $c + \beta x$, so the pooled slope is β and the pooled intercept lies between the group intercepts. Without a condition ensuring a linear pooled conditional median, no such coefficientwise conclusion is valid.

Key Takeaway

The ordering of the two intercepts and slopes is not enough. In particular, the pooled slope can be negative. Interval conclusions require an additional structural condition on the pooled conditional median, not just on the two separate LAD fits.

5 Problem 5: When do no-intercept OLS and LAD have the same slope?

Problem Statement

We fit no-intercept OLS and LAD regressions of y on x and observe that the two slope estimates are the same: $\widehat{b}_{\text{OLS}} = \widehat{b}_{\text{LAD}}$.

(a) Characterise exactly when this happens.

(b) Does it imply that the residuals are symmetric?

(c) Does it imply that the through-origin model is correct?

Required Theory

For no-intercept OLS,

$$\widehat{b}_{\text{OLS}} = \frac{\sum_i x_i y_i}{\sum_i x_i^2},$$

provided $\sum_i x_i^2 > 0$. Its residuals $e_i = y_i - b x_i$ satisfy the normal equation $\sum_i x_i e_i = 0$.

For no-intercept LAD, the objective is $Q_0(b) = \sum_i |y_i - b x_i|$. Its one-dimensional KKT condition is that there exist s_i with $s_i = \text{sign}(y_i - b x_i)$ for nonzero residuals, $s_i \in [-1, 1]$ for zero residuals, and

$$\sum_i x_i s_i = 0.$$

Equivalently, when $x_i \neq 0$,

$$Q_0(b) = \sum_{i: x_i \neq 0} |x_i| \left| \frac{y_i}{x_i} - b \right| + \sum_{i: x_i = 0} |y_i|,$$

so LAD minimizes a weighted absolute-deviation criterion for the ratios y_i/x_i, with weights $|x_i|$. The weighted-median condition in the problem sheet characterizes its minimizers.

Solution

(a) Exact characterization.

Let $b_0 = \widehat{b}_{\text{OLS}}$. The two estimates coincide if and only if b_0 also satisfies the LAD KKT condition. Thus the exact necessary and sufficient condition is

$$\boxed{\begin{array}{c} \text{There exist } s_i \text{ such that } s_i = \text{sign}(y_i - b_0 x_i) \text{ if } y_i - b_0 x_i \neq 0, \\ s_i \in [-1, 1] \text{ if } y_i - b_0 x_i = 0, \qquad \displaystyle\sum_i x_i s_i = 0. \end{array}}$$

This is not merely a sufficient condition: it is exactly the one-dimensional subgradient condition for b_0 to minimize the convex LAD objective.

The weighted-median formulation gives the same characterization in a more interpretable form: the OLS slope must also be an $|x_i|$-weighted median of the ratios y_i/x_i for all observations with $x_i \neq 0$. OLS uses a weighted mean-type equation, whereas LAD uses a weighted median condition; equality occurs when the OLS solution also satisfies the median balance.

(b) It does not imply symmetric residuals.

Take

$$x = (1, 2, 3, 4), \qquad y = (2, -1, -1, 3/4).$$

For the through-origin slope $b = 0$, the residuals are exactly the displayed y-values. The OLS normal equation is

$$\sum_i x_i r_i = 1(2) + 2(-1) + 3(-1) + 4(3/4) = 2 - 2 - 3 + 3 = 0,$$

so the OLS slope is 0. The residual signs are $(+, -, -, +)$, and therefore

$$\sum_i x_i \operatorname{sign}(r_i) = 1 - 2 - 3 + 4 = 0.$$

This is the LAD KKT condition, so $b = 0$ is also an LAD slope. Yet the positive residual magnitudes are 2 and 3/4, while the negative magnitudes are 1 and 1; the residuals are not symmetric about zero.

(c) It does not establish that the through-origin model is correct.

The equality is a statement about two optimization criteria evaluated on one sample. It says that the sample's OLS weighted-mean equation and LAD weighted-median condition happen to select the same slope. It does not prove that the conditional mean is bX, that the conditional median is bX, or that there is no intercept or omitted structure. Model correctness is a population statement and cannot be inferred from this numerical coincidence alone.

Key Takeaway

The OLS and LAD slopes coincide exactly when the OLS slope also satisfies the weighted-median/KKT condition. This coincidence neither implies symmetric residuals nor validates the through-origin model.

6 Problem 6: OLS and LAD with intercept: consequences of exact agreement

Problem Statement

We fit OLS and LAD regressions of y on x with intercept, and observe that both the intercepts and the slopes coincide.

(a) What does this tell us about the residuals?

(b) Does it imply symmetric errors?

(c) In the location–scale model $Y = \alpha + \beta X + (\gamma + \delta X)\varepsilon$, when are the OLS and LAD targets equal?

(d) How likely is exact equality in a sample?

Required Theory

For OLS with intercept, the residuals $e_i = y_i - \widehat{a} - \widehat{b}x_i$ obey

$$\sum_i e_i = 0, \qquad \sum_i x_i e_i = 0.$$

For LAD, the KKT certificate requires values s_i such that

$$s_i = \operatorname{sign}(e_i)\ (e_i \neq 0), \qquad s_i \in [-1, 1]\ (e_i = 0), \qquad \sum_i s_i = 0, \qquad \sum_i x_i s_i = 0.$$

Thus coincident coefficient estimates mean that the OLS residual vector also satisfies the LAD sign-balance conditions.

In the population, the supplied conditional-median result says that a linear conditional median is the population LAD target; the analogous conditional-mean statement says that a linear conditional mean is the population OLS target. A location–scale model therefore allows us to compare its conditional mean and conditional median directly.

Solution

(a) What equality tells us about residuals.

Let the common fitted line be $\widehat{a} + \widehat{b}x$. The OLS normal equations tell us that the signed residual magnitudes balance in the two linear moment equations:

$$\sum_i e_i = 0, \qquad \sum_i x_i e_i = 0.$$

Because the same line is also LAD-optimal, there must additionally be a certificate satisfying

$$\sum_i s_i = 0, \qquad \sum_i x_i s_i = 0,$$

where the certificate is fixed to the residual sign for nonzero residuals and can be chosen in $[-1, 1]$ for zero residuals. In particular, the positive and negative residual counts cannot exceed half the sample size (Problem 8). These are exact balance conditions; they do not state that positive and negative residuals have matching magnitudes or matching distributions.

(b) No symmetry conclusion follows.

For example, take

$$x = (0, 1, 2, 3), \qquad y = \left(\frac{4}{3}, -1, -2, \frac{5}{3}\right).$$

Consider the line $y = 0$. Its residuals are the displayed y_i's. First,

$$\sum_i e_i = \frac{4}{3} - 1 - 2 + \frac{5}{3} = 0,$$

and

$$\sum_i x_i e_i = 0 - 1 - 4 + 5 = 0.$$

Thus the line is the OLS fit. Its residual signs are $(+, -, -, +)$, for which

$$\sum_i s_i = 1 - 1 - 1 + 1 = 0, \qquad \sum_i x_i s_i = 0 - 1 - 2 + 3 = 0.$$

The same line therefore satisfies the LAD KKT conditions. But the positive residual magnitudes are $4/3$ and $5/3$, while the negative magnitudes are 1 and 2. The residuals are not symmetric about zero. Hence exact agreement does not imply symmetric errors.

(c) Equality of population targets in the location–scale model.

Write

$$Y = \alpha + \beta X + s(X)\varepsilon, \qquad s(X) = \gamma + \delta X.$$

Use the usual location–scale interpretation that ε is independent of X (or, more generally, has a conditional mean and a conditional median that do not depend on X). Assume the scale is nonnegative on the predictor range, $\mathbb{E}|\varepsilon| < \infty$, and $\mathbb{E}(\varepsilon^2) < \infty$ when using the OLS target. Then

$$\mathbb{E}(Y \mid X = x) = \alpha + \beta x + (\gamma + \delta x)\mathbb{E}(\varepsilon).$$

If m_ε is a median of ε, the conditional median is

$$\operatorname{med}(Y \mid X = x) = \alpha + \beta x + (\gamma + \delta x)m_\varepsilon.$$

The conditional mean and median are therefore both affine in x. Their coefficient pairs are

$$(a_{\mathrm{OLS}}, b_{\mathrm{OLS}}) = (\alpha + \gamma\mathbb{E}\varepsilon,\ \beta + \delta\mathbb{E}\varepsilon),$$
$$(a_{\mathrm{LAD}}, b_{\mathrm{LAD}}) = (\alpha + \gamma m_\varepsilon,\ \beta + \delta m_\varepsilon).$$

Consequently, the targets coincide whenever

$$\boxed{\mathbb{E}(\varepsilon) = m_\varepsilon.}$$

The common special case is $\mathbb{E}\varepsilon = 0$ and $\operatorname{med}(\varepsilon) = 0$. Symmetry about zero is sufficient for this equality when the moments exist, but it is not necessary. More generally, if $s(X)$ vanishes on the entire predictor support, the disturbance has no effect and the targets coincide regardless of its mean and median; apart from such degeneracies, equality requires the mean and median shifts to agree.

If $s(x)$ can be negative, multiplication reverses the order of the error distribution, but a median still transforms to $s(x)m_\varepsilon$ for a nonzero scalar $s(x)$; the displayed formula remains valid after accounting for the sign. The nonnegative-scale assumption is the usual interpretation of a scale model.

(d) How likely is exact equality in a sample?

Exact equality requires the OLS residual signs to satisfy the LAD KKT equations in addition to the OLS moment equations. If the predictor values are in a generic fixed configuration for which no sign vector $s_i \in \{-1, +1\}$ satisfies

$$\sum_i s_i = 0, \qquad \sum_i x_i s_i = 0,$$

then, under a continuous response distribution (so exact zero residuals have probability zero), OLS and LAD cannot coincide. For such designs exact equality is a probability-zero event. However, this is not a universal statement: a specially balanced design can admit a sign pattern, and then the residuals can fall in a nonempty region of the OLS residual subspace with that sign pattern. For example, at $x = (0, 1, 2, 3)$ the sign vector $(+1, -1, -1, +1)$ satisfies both equations. Small perturbations of the residual vector $(1, -1, -1, 1)$ within the OLS residual subspace preserve these signs, so a continuous error distribution can assign this event positive probability. Thus exact equality is generally restrictive, but its probability depends on the design and the error distribution; it is not always zero.

Key Takeaway

Coincident OLS and LAD lines imply both moment balance and sign balance of the residuals. They do not imply symmetry. At the population level, equality is governed by equality of the error mean and median in the location–scale model.

7 Problem 7: Special LAD cases and a numerical no-intercept example

Problem Statement

(a) Show that LAD with only an intercept gives the sample median.

(b) Show that LAD with intercept and a binary predictor $x \in \{0, 1\}$ gives the two group medians.

(c) Show that no-intercept LAD gives an $|x|$-weighted median of y_i/x_i.

(d) Compute the no-intercept LAD and OLS estimates for the data

$$(1, 1), \quad (2, 5), \quad (3, 3), \quad (4, 9).$$

Required Theory

The unweighted absolute-deviation function $g(a) = \sum_i |y_i - a|$ is minimized at a sample median. A median m satisfies

$$\#\{i : y_i < m\} \le \frac{n}{2} \le \#\{i : y_i \le m\}.$$

For positive weights w_i, the supplied weighted-median result says that b minimizes $\sum_i w_i|z_i - b|$ exactly when

$$\sum_{i:z_i<b} w_i \le \frac{1}{2}\sum_i w_i \le \sum_{i:z_i\le b} w_i.$$

The weighted median is therefore a direct consequence of the subgradient balance for absolute loss.

Solution

(a) LAD with an intercept only gives the sample median.

With no predictor, the fitted value is a constant a and the objective is

$$g(a) = \sum_{i=1}^{n} |y_i - a|.$$

For $y_i \neq a$, a subgradient of $|y_i - a|$ with respect to a is -1 if $y_i > a$ and $+1$ if $y_i < a$. At a tie $y_i = a$, the subgradient is any value in $[-1, 1]$. The necessary and sufficient convex optimality condition is that zero belongs to the total subgradient. This is equivalent to having no more than half the observations strictly below a and no more than half strictly above it. That is precisely the sample-median condition.

(b) A binary predictor gives the two group medians.

The model is $a + bx$. For $x = 0$, the fitted value is a; for $x = 1$, it is $a + b$. Split the objective by group:

$$Q(a, b) = \sum_{i:x_i=0} |y_i - a| + \sum_{i:x_i=1} |y_i - (a + b)|.$$

The transformation from (a, b) to $(m_0, m_1) = (a, a + b)$ is one-to-one, with inverse $a = m_0$, $b = m_1 - m_0$. Thus the minimization separates:

$$\widehat{m}_0 \in \text{med}\{y_i : x_i = 0\}, \qquad \widehat{m}_1 \in \text{med}\{y_i : x_i = 1\}.$$

The resulting coefficients are $\widehat{a} = \widehat{m}_0$ and $\widehat{b} = \widehat{m}_1 - \widehat{m}_0$.

(c) No-intercept LAD gives an $|x|$-weighted median.

For $x_i \neq 0$,

$$|y_i - bx_i| = |x_i| \left| \frac{y_i}{x_i} - b \right|.$$

Terms with $x_i = 0$ equal $|y_i|$ and do not depend on b. Therefore

$$Q_0(b) = \text{constant} + \sum_{i:x_i \neq 0} |x_i| \left| \frac{y_i}{x_i} - b \right|.$$

This is a weighted absolute-deviation criterion with $z_i = y_i/x_i$ and $w_i = |x_i|$. Hence every LAD slope is an $|x_i|$-weighted median of the ratios y_i/x_i.

(d) Numerical estimates.

For the four observations, the ratios and weights are

(x_i, y_i)	$(1,1)$	$(2,5)$	$(3,3)$	$(4,9)$
y_i/x_i	1	5/2	1	9/4
$\lvert x_i \rvert$	1	2	3	4

Sort the ratios:

$$1 \text{ (weight } 1+3=4), \qquad \frac{9}{4} \text{ (weight 4)}, \qquad \frac{5}{2} \text{ (weight 2)}.$$

The total weight is $W = 1+2+3+4 = 10$, so half the weight is 5. The cumulative weight strictly below 9/4 is 4, and the cumulative weight at or below 9/4 is 8. Thus 9/4 is the weighted median, and

$$\boxed{\widehat{b}_{\text{LAD}} = \frac{9}{4} = 2.25.}$$

For no-intercept OLS,

$$\widehat{b}_{\text{OLS}} = \frac{\sum_i x_i y_i}{\sum_i x_i^2} = \frac{1+10+9+36}{1+4+9+16} = \frac{56}{30} = \boxed{\frac{28}{15} \approx 1.8667.}$$

Key Takeaway

Median regression is not a mysterious special algorithm: it is the minimization of absolute deviations. With a no-intercept predictor, each ratio y_i/x_i receives weight $|x_i|$.

8 Problem 8: Residual-sign counts for LAD and quantile regression

Problem Statement

Let $(\widehat{a}, \widehat{b})$ be any LAD fit with intercept.

(a) Show that $n_- \leq n/2 \leq n_- + n_0$ (equivalently, $n_+ \leq n/2$ and $n_- \leq n/2$).

(b) For quantile regression at level $\tau \in (0,1)$, with loss $\rho_\tau(u) = u(\tau - \mathbf{1}\{u < 0\})$, show that $n_- \leq n\tau \leq n_- + n_0$.

(c) Converse: if a line has $n_+ \leq n/2$ and $n_- \leq n/2$, must it be an LAD line?

Required Theory

For an intercept-and-slope LAD fit, the intercept KKT equation is $\sum_i s_i = 0$, where $s_i = 1$ for positive residuals, $s_i = -1$ for negative residuals, and $s_i \in [-1, 1]$ at zero residuals. Let n_+, n_-, n_0 denote the corresponding counts.

For quantile loss,

$$\rho_\tau(u) = u(\tau - \mathbf{1}\{u < 0\}),$$

the derivative with respect to u is τ for $u > 0$ and $\tau - 1$ for $u < 0$. At $u = 0$, the subgradient interval is $[\tau - 1, \tau]$. With an intercept, the sum of these subgradients must contain zero.

Solution

(a) LAD sign counts.

The intercept KKT equation can be written

$$0 = n_+ - n_- + \sum_{i:r_i=0} s_i, \qquad -n_0 \le \sum_{i:r_i=0} s_i \le n_0.$$

The lower bound gives

$$0 \ge n_+ - n_- - n_0 \quad \Longleftrightarrow \quad n_+ \le n_- + n_0.$$

The upper bound gives

$$0 \le n_+ - n_- + n_0 \quad \Longleftrightarrow \quad n_- \le n_+ + n_0.$$

Since $n = n_+ + n_- + n_0$, the first inequality is equivalent to $2n_+ \le n$, and the second to $2n_- \le n$. Hence

$$\boxed{n_+ \le n/2, \qquad n_- \le n/2.}$$

Equivalently, $n_- \le n/2 \le n_- + n_0$.

(b) Quantile regression.

For the intercept-only subgradient balance (which is necessary for any intercept-and-slope quantile fit), the nonzero residuals contribute

$$n_+\tau + n_-(\tau - 1).$$

The n_0 zero residuals contribute some number in the interval $[n_0(\tau - 1), n_0\tau]$. Therefore zero can be achieved exactly when

$$n_+\tau + n_-(\tau - 1) + n_0(\tau - 1) \le 0$$

and

$$n_+\tau + n_-(\tau - 1) + n_0\tau \ge 0.$$

Using $n = n_+ + n_- + n_0$, the first inequality becomes

$$n\tau - n_- - n_0 \le 0,$$

and the second becomes

$$n\tau - n_- \ge 0.$$

Thus

$$\boxed{n_- \le n\tau \le n_- + n_0.}$$

For $\tau = 1/2$, this reduces to the LAD median count condition.

(c) The converse is false.

The sign counts only verify the intercept balance; they do not verify the slope balance. Consider four observations with predictor values $x = (0, 1, 2, 3)$ and responses $y = (1, 1, -1, -1)$. Examine the horizontal line $a = 0, b = 0$. It has two positive and two negative residuals, so

$$n_+ = n_- = 2 = n/2.$$

However, since none of the residuals is zero, the certificate is forced to be $s = (1, 1, -1, -1)$. Its slope sum is

$$\sum_i x_i s_i = 0 + 1 - 2 - 3 = -4 \neq 0.$$

The slope KKT condition fails. Therefore the line is not an LAD fit, despite satisfying the sign-count bounds.

Key Takeaway

The count inequalities are necessary because the intercept must balance positive and negative residual signs. They are not sufficient: a fitted line must also balance the predictor-weighted signs.

9 Problem 9: The entire LAD solution set for unit-square data

Problem Statement

(a) For the unit-square data $(0, 0)$, $(0, 1)$, $(1, 0)$, $(1, 1)$, find the entire set of LAD solutions (a, b).

(b) Relate the extreme points of this set to Problem 1.

(c) Give a simple sufficient condition for uniqueness.

Required Theory

The LAD objective is a convex piecewise-linear function. Its minimizer set is therefore convex; in this example it can be described exactly by inequalities. The KKT certificate characterizes the minimizers, while the supplied uniqueness result gives a simple sufficient condition: two observations on the fitted line at distinct predictor values, with strictly interior certificate values at those zero residuals, force uniqueness.

Solution

(a) Find the whole set.

For the four points,

$$Q(a, b) = |a| + |1 - a| + |a + b| + |1 - a - b|.$$

The elementary inequality

$$|z| + |1 - z| \geq 1$$

holds for every z, with equality if and only if $z \in [0, 1]$. Apply it first to $z = a$ and then to $z = a + b$:

$$Q(a, b) \geq 1 + 1 = 2.$$

Equality holds precisely when both a and $a+b$ belong to $[0,1]$. Thus the minimum value is 2 and the complete minimizer set is

$$\boxed{\mathcal{M} = \{(a,b) : 0 \le a \le 1,\ 0 \le a+b \le 1\}.}$$

Equivalently, $0 \le a \le 1$ and $-a \le b \le 1-a$.

(b) Extreme points and Problem 1.

The feasible set is a parallelogram in the (a,b) plane. Its four extreme points are

$$(0,0), \quad (0,1), \quad (1,-1), \quad (1,0).$$

The corresponding lines are $y=0$, $y=x$, $y=1-x$, and $y=1$. Each passes through two square vertices with different x-values:

(a,b)	two data points on the line
$(0,0)$	$(0,0),(1,0)$
$(0,1)$	$(0,0),(1,1)$
$(1,-1)$	$(0,1),(1,0)$
$(1,0)$	$(0,1),(1,1)$

This illustrates Problem 1(b): an extreme minimizer can be selected that passes through two observations at distinct predictor values. It does not say every point of the minimizer parallelogram has that property.

(c) A sufficient condition for uniqueness.

A direct sufficient condition is the supplied Tool B: if at a proposed LAD fit there are two zero residuals at distinct x-values and a valid KKT certificate assigns both corresponding observations values strictly inside $(-1,1)$, then the solution is unique. The unit-square example does not satisfy this condition at its extreme points in the displayed natural certificates, and indeed its minimizer set is not a singleton.

Key Takeaway

The complete solution set can be a region rather than a single line. Extreme points of this region correspond to supporting lines through two observations at different predictor values.

10 Problem 10: Perturbing responses and the effect of an extreme predictor

Problem Statement

Let $(\widehat{a}, \widehat{b})$ be an LAD fit.

(a) Move any y_i vertically without changing the sign of its residual (points on the fitted line stay put). Show that $(\widehat{a}, \widehat{b})$ remains an LAD fit. What happens to OLS?

(b) Show that a single point with extreme x can make the LAD slope any real number.

Required Theory

The KKT certificate for LAD depends on the signs of nonzero residuals and on subgradient values at zero residuals. The magnitude of a nonzero residual does not enter the certificate. OLS, in contrast, depends on the actual residual magnitudes through the two normal equations.

Solution

(a) LAD is unchanged as long as residual signs are unchanged.

Start from an LAD fit with residuals $r_i = y_i - \widehat{a} - \widehat{b}x_i$ and a valid certificate (s_i). Replace the response values by y_i^*, subject to the conditions that each nonzero residual keeps its sign and every point originally on the fitted line remains on it. The new residuals are

$$r_i^* = y_i^* - \widehat{a} - \widehat{b}x_i.$$

For a residual that was nonzero, $\operatorname{sign}(r_i^*) = \operatorname{sign}(r_i)$, so its certificate value remains valid. A residual that was zero remains zero, so its old value $s_i \in [-1, 1]$ remains valid. Consequently the same certificate satisfies

$$\sum_i s_i = 0, \qquad \sum_i x_i s_i = 0.$$

The KKT conditions still hold, and $(\widehat{a}, \widehat{b})$ remains an LAD fit.

OLS is generally not invariant. If the vertical changes are $d_i = y_i^* - y_i$, the OLS normal equations at the old line become

$$\sum_i e_i^* = \sum_i e_i + \sum_i d_i = \sum_i d_i, \qquad \sum_i x_i e_i^* = \sum_i x_i e_i + \sum_i x_i d_i = \sum_i x_i d_i.$$

Unless both added sums happen to be zero, the old coefficients cease to satisfy the OLS normal equations. Hence OLS will generally change.

(b) One extreme-x observation can support any prescribed LAD slope.

Fix an arbitrary real number B. Begin with two observations at the same predictor value:

$$(0, -1), \qquad (0, 1).$$

Add one point at an extreme predictor value $M \neq 0$, with response BM. Consider the line $y = Bx$. Its residuals at the first two points are -1 and 1, and its residual at the added point is zero. Its total absolute loss is therefore 2.

For any competing line $y = a + bx$, the loss contributed by just the first two observations is

$$|-1 - a| + |1 - a| \geq 2,$$

while the added point contributes a nonnegative amount. Thus no line can have total loss below 2, and the line $y = Bx$ attains this lower bound. It is an LAD fit, with slope exactly B. Since B was arbitrary, a single point at an extreme x can make any real number an LAD slope. The combined sample has predictor variation because $M \neq 0$.

Key Takeaway

LAD's KKT conditions are sign-based, so vertical changes that preserve residual signs leave the fit optimal. A point at a sufficiently distinctive predictor value can also support an arbitrarily chosen slope.

11 Problem 11: Adding observations on or around a fitted line

Problem Statement

Let $(\widehat{a}, \widehat{b})$ be an LAD fit, with fitted value $\widehat{y}_0 = \widehat{a} + \widehat{b}x_0$ at some x_0.

(a) Add a point on the fitted line, $(x_0, \widehat{y}_0)$. Does the fit change?

(b) Add a symmetric pair $(x_0, \widehat{y}_0 + d)$ and $(x_0, \widehat{y}_0 - d)$, $d > 0$. Does the fit change?

(c) Can adding a single point off the line change the fit?

Required Theory

If an old objective $Q(a, b)$ has minimum m, adding an observation adds a nonnegative term to it. If the proposed old minimizer makes that new term as small as possible, it remains a minimizer. The KKT certificate gives a second way to see the result: an added zero residual can be assigned certificate value zero, and a symmetric pair at the same x can be assigned opposite certificate values, so their contributions to both KKT equations cancel.

Solution

(a) Adding a point on the fitted line does not remove the fit.

The new residual at $(x_0, \widehat{y}_0)$ is zero. The old objective is at least m for every line, so the augmented objective is at least m. At the old fitted line the new point contributes zero and the objective remains m. Therefore the old fit remains LAD-optimal. It may cease to be unique only if the augmented geometry permits additional minimizers; the statement guaranteed here is that the fit remains a minimizer.

In certificate terms, assign the added zero residual $s_0 = 0$. It changes neither $\sum s_i$ nor $\sum x_i s_i$.

(b) Adding a symmetric pair also preserves the fit.

Let a competing line have fitted value $a + bx_0 = \widehat{y}_0 + u$ at x_0. The two new absolute residuals sum to

$$|d - u| + | - d - u| = |d - u| + |d + u| \geq 2d,$$

by the triangle inequality. The old fitted line has $u = 0$, so it contributes exactly $2d$. For any competing line the old observations contribute at least m and the pair contributes at least $2d$. Hence the augmented objective is at least $m + 2d$, and the old fit attains that lower bound.

Equivalently, assign the two new residuals certificate values $+1$ and -1. Their intercept contributions sum to zero and their slope contributions are $x_0 - x_0 = 0$.

(c) A single point off the line can change the fit.

Start with the two observations $(0, 0)$ and $(1, 0)$. The line $y = 0$ fits them exactly and is the unique zero-loss LAD line. Add the point $(2, 10)$. The old line now has objective 10. The line through $(0, 0)$ and $(2, 10)$ is $y = 5x$; its residuals at the three observations are

$$0, \quad -5, \quad 0,$$

so its objective is 5. Therefore the old line is no longer optimal. A single observation off the line can change the fit.

Key Takeaway

A point on the line or a symmetric vertical pair at the same predictor cannot make the existing LAD fit worse than any competitor in the relevant added-loss comparison. A single unbalanced off-line point can change the solution.

12 Problem 12: A full-interaction LAD model is equivalent to two separate fits

Problem Statement

Let $g_i \in \{0, 1\}$ indicate the subpopulation. Fit by LAD the full-interaction model

$$y = a + bx + cg + d(g \cdot x).$$

(a) Show that this reproduces the two separate subpopulation LAD fits.

(b) Contrast this with pooling (Problems 3 and 4).

Required Theory

The LAD objective is the sum of absolute residuals. A binary group indicator with a full interaction allows both the intercept and the slope to differ by group. The parameter transformation between (a, b, c, d) and the two group-specific coefficient pairs is one-to-one:

$$(a_0, b_0) = (a, b), \qquad (a_1, b_1) = (a + c, b + d).$$

Because the absolute-loss sum separates by group, unconstrained minimization in these four equivalent parameters separates into two independent LAD problems.

Solution

(a) Derive the two group lines and separate the objective.

For $g_i = 0$, the fitted value is

$$\widehat{y}_i = a + bx_i.$$

For $g_i = 1$, it is

$$\widehat{y}_i = (a + c) + (b + d)x_i.$$

Let S_0 and S_1 be the two subpopulations. The total LAD objective is

$$\begin{aligned} Q(a, b, c, d) = &\sum_{i \in S_0} |y_i - a - bx_i| \\ &+ \sum_{i \in S_1} |y_i - (a + c) - (b + d)x_i|. \end{aligned}$$

Now define the group coefficients

$$a_0 = a, \quad b_0 = b, \quad a_1 = a + c, \quad b_1 = b + d.$$

The inverse transformation is

$$a = a_0, \quad b = b_0, \quad c = a_1 - a_0, \quad d = b_1 - b_0.$$

Thus optimizing over (a, b, c, d) is exactly equivalent to optimizing over the four unrestricted group coefficients (a_0, b_0, a_1, b_1). The objective becomes

$$Q_0(a_0, b_0) + Q_1(a_1, b_1),$$

so its minimization separates. Each group receives its own LAD intercept and slope, exactly as in the two separate fits.

(b) Contrast with ordinary pooling.

Ordinary pooling without the group indicator uses one common line $a + bx$ for both groups. It imposes

$$a_0 = a_1, \qquad b_0 = b_1,$$

which are restrictions not present in the full-interaction model. Problems 3 and 4 show that the best common line can have coefficients outside the range of the groupwise coefficients, including a negative pooled slope even when both groupwise slopes are positive. The full-interaction model avoids this forced compromise by allowing separate intercepts and slopes.

Key Takeaway

A group indicator alone allows separate intercepts; adding its interaction with x also allows separate slopes. With both terms included, LAD exactly reproduces the two independent groupwise fits.

13 Problem 13: Equivariance of LAD under affine transformations

Problem Statement

Let $(\widehat{a}, \widehat{b})$ be an LAD fit of y on x (with intercept).

(a) Find the LAD fit of $y^* = cy + a_0 + b_0 x$ on x, where $c \neq 0$.

(b) Find the LAD fit of y on $x^* = px + q$, where $p \neq 0$.

(c) Population: if $\text{med}(\log Y \mid X) = a + bX$, what is $\text{med}(Y \mid X)$? Is the analogous statement true for conditional means?

(d) Is $\text{med}(h(Y)) = h(\text{med}(Y))$ for every function h?

Required Theory

An LAD fit minimizes the sum of absolute residuals. Multiplying every residual by a nonzero constant multiplies the objective by the positive constant $|c|$ and therefore preserves the minimizers after the corresponding coefficient transformation. A strictly increasing function maps medians to medians because it preserves order. Conditional expectations do not generally commute with nonlinear transformations.

Solution

(a) Transforming the response.

The original fitted line is $\widehat{a} + \widehat{b}x$. Under

$$y^* = cy + a_0 + b_0 x,$$

a natural transformed line is

$$\widehat{y}^* = c(\widehat{a} + \widehat{b}x) + a_0 + b_0 x.$$

Its intercept and slope are

$$\boxed{\widehat{a}^* = c\widehat{a} + a_0, \qquad \widehat{b}^* = c\widehat{b} + b_0.}$$

To verify optimality, for any candidate (a, b) in the transformed problem subtract the fixed affine part and divide by c:

$$\sum_i |y_i^* - a - bx_i| = |c| \sum_i \left| y_i - \frac{a - a_0}{c} - \frac{b - b_0}{c} x_i \right|.$$

The map $(a, b) \mapsto ((a - a_0)/c, (b - b_0)/c)$ is one-to-one. Hence minimizing the transformed objective is equivalent to the original LAD problem, proving the stated coefficients.

(b) Transforming the predictor.

Since $x^* = px + q$, we have $x = (x^* - q)/p$. The original line becomes

$$\widehat{a} + \widehat{b}\frac{x^* - q}{p} = \left(\widehat{a} - \frac{q\widehat{b}}{p}\right) + \frac{\widehat{b}}{p} x^*.$$

Thus the transformed intercept and slope are

$$\boxed{\widehat{a}^* = \widehat{a} - \frac{q\widehat{b}}{p}, \qquad \widehat{b}^* = \frac{\widehat{b}}{p}.}$$

This is optimal because the transformation between candidate lines in x and candidate lines in x^* is one-to-one, and the fitted values at every observation are unchanged.

(c) Conditional median of Y.

Assume $Y > 0$ so that $\log Y$ is defined. The exponential function is strictly increasing. Therefore it preserves the median ordering, and

$$\operatorname{med}(Y \mid X) = \exp\{\operatorname{med}(\log Y \mid X)\} = \boxed{e^{a+bX}}.$$

The analogous statement for means is false in general:

$$\mathbb{E}(Y \mid X) \neq \exp\{\mathbb{E}(\log Y \mid X)\}$$

without further distributional assumptions. A simple reason is that expectation is an average of values, whereas exponentiation is nonlinear; the expectation of an exponential is not generally the exponential of an expectation.

(d) Not for every function.

The median transformation identity is guaranteed for monotone transformations (with the usual care for non-unique medians and decreasing transformations). It is false for arbitrary non-monotone h. For example, let Y take values -1 and 1 with probability $1/2$ each. Then 0 is a median of Y. With $h(y) = y^2$, however, $h(Y) = 1$ with probability one, so its median is 1, whereas $h(0) = 0$. Thus

$$\operatorname{med}(h(Y)) = 1 \neq 0 = h(\operatorname{med}(Y)).$$

Key Takeaway

LAD is equivariant under nonzero affine transformations of the response and invertible affine transformations of the predictor. The median commutes with monotone transformations, not arbitrary functions; the mean does not generally commute with nonlinear transformations.

14 Problem 14: Medians of mixtures and conditional median pooling

Problem Statement

(a) Let $F = \pi F_1 + (1-\pi)F_2$ with $\pi \in (0,1)$, and let $m_1 \le m_2$ be medians of F_1 and F_2 respectively. Show that F has a median in $[m_1, m_2]$.

(b) In the two-group model (group independent of X), each group has a linear conditional median. Must the pooled conditional median $\text{med}(Y \mid X = x)$ be linear in x? Must it be the π-weighted average of the group medians?

Required Theory

A number m is a median of a distribution function F if

$$F(m-) \le \frac{1}{2} \le F(m),$$

where $F(m-)$ is the left limit. A mixture distribution has CDF $F = \pi F_1 + (1-\pi)F_2$. Conditional on $X = x$, if group membership is independent of X, the mixture weights remain π and $1-\pi$; the pooled conditional CDF is the same weighted mixture of the two groupwise conditional CDFs.

Solution

(a) A mixture has a median between the two component medians.

Because m_1 is a median of F_1,

$$F_1(m_1-) \le \frac{1}{2} \le F_1(m_1).$$

Because $m_1 \le m_2$ and CDFs are nondecreasing,

$$F_2(m_1-) \le F_2(m_2-) \le \frac{1}{2}.$$

Therefore

$$F(m_1-) = \pi F_1(m_1-) + (1-\pi)F_2(m_1-) \le \frac{1}{2}.$$

Similarly, monotonicity and the fact that m_1 is a median imply

$$F_1(m_2) \ge F_1(m_1) \ge \frac{1}{2},$$

while $F_2(m_2) \ge 1/2$. Hence

$$F(m_2) = \pi F_1(m_2) + (1-\pi)F_2(m_2) \ge \frac{1}{2}.$$

The CDF must cross the median level somewhere between these two points. More formally, the lower median $m = \inf\{y : F(y) \ge 1/2\}$ lies in $[m_1, m_2]$. Thus the mixture has at least one median in that interval.

(b) The pooled conditional median need not be linear or a weighted average.

The mixture result applies separately at every fixed x: any pooled conditional median lies between the two groupwise conditional medians at that x. But the point between the two curves can vary nonlinearly with x.

Here is a concrete example with group membership independent of X and equal group probabilities. Let

$$Y \mid (X = x, G = 1) \sim \text{Uniform}(-1, 1),$$

so the group-1 conditional median is $m_1(x) = 0$. Let

$$Y \mid (X = x, G = 2) \sim \text{Uniform}(x - (1 + x^2),\ x + (1 + x^2)),$$

so the group-2 conditional median is $m_2(x) = x$. Take $\pi = 1/2$. For the values of x considered below, the candidate median lies in the interior of both uniform supports. At such a point y, the two conditional CDFs are

$$F_1(y \mid x) = \frac{1}{2} + \frac{y}{2}, \qquad F_2(y \mid x) = \frac{1}{2} + \frac{y - x}{2(1 + x^2)}.$$

The pooled median solves

$$\frac{1}{2}F_1(y \mid x) + \frac{1}{2}F_2(y \mid x) = \frac{1}{2}.$$

Substituting the CDFs and simplifying gives

$$y + \frac{y - x}{1 + x^2} = 0 \quad \Longrightarrow \quad \boxed{m_{\text{pool}}(x) = \frac{x}{2 + x^2}.}$$

This is not an affine function of x: for instance, its values at $x = 0, 1, 2$ are $0, 1/3, 1/3$, which cannot lie on one straight line. It is also not the π-weighted average of the group medians, which would be $x/2$ in this equal-weight example. Therefore neither assertion is generally true.

The example also clarifies the point of part (a): at each x, $x/(2 + x^2)$ lies between 0 and x for $x \geq 0$, but the way it moves between the two group median curves is nonlinear.

Key Takeaway

Independence of group membership and X keeps the mixture weights fixed conditional on X, but a mixture of two linear conditional-median models need not have a linear conditional median. The pooled median is generally not the weighted average of the component medians.

15 Problem 15: Simpson-type reversal: positive within-group slopes and a negative pooled slope

Problem Statement

(a) Construct two subpopulations with the same positive LAD slope for which the pooled LAD slope is negative.

(b) Show that adding a group indicator (common slope, separate intercepts) recovers the correct slope in your example.

Required Theory

A common-slope model with a group indicator is

$$y = a + bx + cg.$$

It permits the group intercepts to differ while imposing one shared slope. The objective is the sum of the two groupwise absolute-deviation objectives, with a common b. If each group lies exactly on a line with the same slope but different intercepts, the common-slope/group-intercept model can fit both groups exactly.

Ordinary pooling without g forces one line to represent both groups. If the predictor ranges and intercepts differ substantially, the between-group arrangement can dominate the within-group positive trend. The KKT certificate verifies the pooled LAD solution.

Solution

(a) Construct the reversal.

Take

$$S_1 : \ (0,0), (1,1), \qquad S_2 : \ (10,-100), (11,-99).$$

The first group lies exactly on $y = x$, so its LAD slope is $+1$. The second lies exactly on $y = x - 110$, so its LAD slope is also $+1$. Each two-point group has distinct predictor values and its own exact line is the unique groupwise LAD fit.

Now pool the observations and ignore the group indicator. Consider the line

$$\boxed{\widehat{y} = -9x.}$$

Its residuals in the order $(0,0), (1,1), (10,-100), (11,-99)$ are

$$(0, 10, -10, 0).$$

The two nonzero residual signs are $+1$ at $x = 1$ and -1 at $x = 10$. At the zero residuals $x = 0$ and $x = 11$, choose certificate values $-9/11$ and $9/11$, respectively. Then

$$\sum_i s_i = -\frac{9}{11} + 1 - 1 + \frac{9}{11} = 0,$$

and

$$\sum_i x_i s_i = 0 + 1 - 10 + 11\frac{9}{11} = 0.$$

This is a valid KKT certificate. Since the two zero residuals occur at distinct predictor values and both zero-residual certificate values have absolute value below one, the supplied uniqueness condition shows that $y = -9x$ is the unique pooled LAD line. Its slope is -9, even though both groupwise slopes are $+1$.

(b) Adding a group indicator recovers the common slope.

Fit

$$y = a + bx + cg,$$

where $g = 0$ for S_1 and $g = 1$ for S_2. In group 1, the fitted line is $a + bx$. The exact data require $a = 0$ and $b = 1$. In group 2, the fitted line is $(a + c) + bx$; with $a = 0$ and $b = 1$, taking $c = -110$ fits both observations in group 2 exactly:

$$y = x - 110.$$

The LAD objective is then zero, its smallest possible value. Thus the common-slope model with separate intercepts recovers

$$\boxed{\widehat{b} = 1, \qquad \widehat{a} = 0, \qquad \widehat{c} = -110.}$$

The negative pooled slope was a consequence of omitting the group intercept, not a negative within-group relationship.

Key Takeaway

A pooled regression can reverse the within-group trend when group membership is associated with both predictor ranges and response levels. A group indicator removes the between-group intercept shift and recovers the common positive slope in this construction.

A Compact reference: LAD optimality conditions

For the intercept-and-slope objective

$$Q(a, b) = \sum_i |y_i - a - bx_i|,$$

a candidate (a, b) is LAD-optimal if and only if there are s_i satisfying

$$s_i = \operatorname{sign}(y_i - a - bx_i) \text{ for nonzero residuals}, \quad s_i \in [-1, 1] \text{ for zero residuals},$$

$$\sum_i s_i = 0, \qquad \sum_i x_i s_i = 0.$$

Without an intercept, only the second equation is required. For a weighted median, the condition is

$$W(z < b) \le W/2 \le W(z \le b).$$

B Connection with the attached ASM II lecture notes

The attached notes explain that LAD uses absolute loss, which is less sensitive to large residuals than squared loss, but that its objective is nondifferentiable at zero residuals. They then derive a majorization inequality and show that minimizing the resulting surrogate leads to iteratively reweighted least squares, with weights proportional to $1/|r_i^{(t)}|$ (Lecture VII, pages 274–277 of the supplied notes). This is an optimization route to LAD. The KKT certificate and median tools used throughout the present problem set are stated explicitly in the attached LAD practice-problem sheet.